> *This lab uses **PHP 8.4**, **Laravel 12** and **Visual Studio Code** as the primary editor.*

> *Make sure you have the **SQLite** extension enabled in your `php.ini` file.*

10.7.4 Steps

Here are the steps to complete this lab:

10.7.4.1 Step 1: Create a New Laravel 12 Project

Open a terminal and create a new Laravel project:

```
laravel new complex-relationships
cd complex-relationships
code .
```

You should see a new folder named `complex-relationships` created. Open the project in your favorite code editor, such as Visual Studio Code.

10.7.4.2 Step 2: Configure SQLite Database

Laravel 12 **uses SQLite by default**, so we need to configure it to use **SQLite**.

Open `.env` and update the database settings. Ensure the following lines are present:

```
DB_CONNECTION=sqlite
```

You don’t need to specify `DB_DATABASE` for SQLite, as it will use the default SQLite database file. Unless you want to specify a custom path, you can leave it as is.

10.7.4.3 Step 3: Create Migrations for the Database Schema

We will create migrations for the following tables:

- `users`: Stores user information. We will modify the default migration.
- `profiles`: Stores user profiles (One-to-One relationship with users)
- `posts`: Stores blog posts (One-to-Many relationship with users)
- `tags`: Stores tags for posts (Many-to-Many relationship with posts)
- `post_tag`: Pivot table for Many-to-Many relationship between posts and tags